Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/31/2006
PI Name:	Reasenberg, Robert Ph.D.		
Project Title:	A Galilean laboratory test of the equivalence principle leading to a Space Station experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02138-1516	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Phillips, James D. (Smithsonian Institution)		
Grant/Contract No.:	GSN 4496		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

We propose to test the weak equivalence principle (WEP), which undergirds general relativity and thus all of relativistic astrophysics and cosmology, by testing the universality of free fall (UFF). A null result would enhance our confidence in general relativity as well as the standard cosmology and interpretation of astrophysical data. An apparent violation of the WEP would have profound implications, either showing the existence of a new fundamental force or demonstrating the failure of general relativity.
The Principle of Equivalence Measurement (POEM) will seek possible differences in the interaction of different kinds of matter with a gravitational field. Such differences can be described in terms of Delta g, the differences in the apparent gravitational acceleration of two samples. We expect to achieve sigma(Delta g)/g = 5 x 10^-14 on the ground and sigma(Delta g)/g = 10^-16 at the International Space Station. The latter accuracy is higher than is expected from any

Task Description:	other project except for dedicated free-flying missions. The ground-based experiment owes its sensitivity to two factors: the instrument is sensitive to the full (vertical) gravity of earth, and the principal measurement tool, a laser gauge, has very high precision. The space-based experiment has the same advantages. Its improved sensitivity comes principally from the longer free-fall time available in a micro-g environment. In both versions, the measurement will be made at "room temperature," avoiding the complexity of cryogenics. Thermal noise poses no problem since the Test Mass Assemblies, which are the only components whose gravity-induced differential motions are measured, are isolated from the environment. They are neither constrained in any degree of freedom, nor connected by a spring or fiber, nor subject to (noisy) position feedback. In this regard, POEM is unique among equivalence principle tests. The objective is to test the weak equivalence principle, which undergirds general relativity and thus all of relativistic astrophysics and cosmology. A null result more stringent than previously available would enhance our confidence in general relativity as well as the standard cosmology and interpretation of astrophysical data. An apparent violation of the weak equivalence principle would have profound implications, either showing the existence of a new fundamental force or demonstrating the failure of general relativity. Either an enhanced null result or the discovery of a true equivalence principle violation would provide constraints on the kinds of theories, such as string theories, that could replace general relativity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	At the beginning of the reporting period, we moved the equipment into a larger space that had been renovated for the principle of equivalence measurement (POEM). The larger space was required to accommodate the torsion-bar version of the “bouncer,” which was designed during the previous year. This year, the parts for the torsion-bar bouncer were fabricated, and the torsion-bar bouncer was assembled. Tests have shown that it meets its primary objectives: It returns the vacuum chamber to upward free fall with minimal loss of energy, and it introduces negligible shock and vibration. In both regards, it is a substantial improvement over the original version. A set of equations of motion have been written for the flying vacuum chamber with the torsion-bar bouncer. These have been modified to allow for test-mass launch. The modified equations were integrated and a means developed to allow the resulting ephemeris to be transferred to the motion controller quickly and automatically. The complete motion system has been successfully tested. We demonstrated during the reporting period that the vibration level in the motion system was to large to achieve reliable launch of the test masses, free from spurious motion. We therefore concluded that it was time to implement the long-planned transition from mechanical bearings to air bearings in the motion system. After substantial investigation, a technical approach was selected, and detailed designs prepared. Assembly and testing are expected early in the next reporting period. The principal measuring instrument for POEM is the laser gauge. During the reporting period, we advanced the development of the version of the laser-gauge based on 1550 nm DFB semiconductor lasers. We purchased and set up a high-finesse reference cavity, locked two lasers to it (using different cavity orders), and characterized the behavior of the system. We have thoroughly characterized the temporal response of the lasers' frequency to their control signal. This is a combination of carrier density effects, which occur in times too short to be of concern for the laser gauge, and a thermal response with several effective time constants. Strategies have been devised to compensate for the thermal response, and to construct a servo system that is stable and meets our goals for speed and accuracy. The work on the capacitance gauge was begun last year as an activity of opportunity when we developed a collaboration with Winfield Hill (Rowland Institute for Science at Harvard). The capacitance gauge is essential to the third generation of POEM and will enhance the first generation effort, which is supported by the present grant. Hill’s design for the 20 channel capacitance gauge includes three kinds of boards plus power supplies, computer interface, etc. The electrode-driver boards and analog-to-digital-converter boards have been designed and fabricated; the circuit for the pre-amplifier boards has been designed.
Bibliography Type:	Description: (Last Updated: 05/07/2008)
Abstracts for Journals and Proceedings	Reasenberg RD, Phillips JD. "An Equivalence Principle Test During a 1 s Suborbital Flight." Proceedings of the Conference on Testing the Equivalence Principle in Space and on Ground, Pescara, Italy, January 2006. Proceedings of the Conference on Testing the Equivalence Principle in Space and on Ground, Pescara, Italy, January 2006. , Jan-2006
Articles in Peer-reviewed Journals	Phillips JD, Reasenberg RD. "Tracking frequency laser distance gauge." Review of Scientific Instruments 2005 May;76(6):064501. http://dx.doi.org/10.1063/1.1921408 , May-2005
Books/Book Chapters	Reasenberg RD, Phillips JD. "Technology for Precision Gravity Measurements." in "Lasers, Clocks and Drag-free Control : exploration of relativistic gravity in space." Ed. H. Dittus, C. Lammerzahl, SG Turyshev. Berlin, New York : Springer, 2008. p. 263-284., Jan-2008